

ECN 164 International Finance — Practice Mock Exam A

Questions · Unofficial self-study practice based on Assignment 1, Part I (Chapter 2: quotes, cross rates, bid–ask spreads, triangular arbitrage)

Instructions. 15 short-answer problems, 100 points, suggested time 90 minutes. Show your work. Round exchange rates to 4 decimal places (more when a rate is very small) and money amounts to 2 decimal places. A calculator is allowed.

Name: _____ Score: _____ / 100

1. (5 points) Flipping a quote. On October 2, 2026 the Australian dollar was worth about USD0.6550/AUD and the Canadian dollar about USD0.7250/CAD.

- How many Australian dollars does one U.S. dollar buy (AUD/USD)?
- How many Canadian dollars does one U.S. dollar buy (CAD/USD)?
- Which currency is the “base” currency in each of the original quotes?

2. (5 points) Paying in a foreign currency. You are traveling in South Korea and your hotel costs ₩180,000 (KRW) per night. The exchange rate is KRW1,380/USD.

- What is the cost of one night in dollars?
- A week later the rate is KRW1,395/USD. Did the won appreciate or depreciate against the dollar? What does the same room cost in dollars now?

3. (5 points) Nordic Furniture Imports, Inc. needs to buy 5,000,000 Swedish kronor (SEK) to pay its Swedish supplier. Its banker quotes bid–ask rates of SEK9.4150–9.4200/USD. What will be the dollar cost of the SEK5,000,000?

4. (7 points) Rates of change. The Australian dollar was USD0.6620/AUD on September 25, 2026 and USD0.6550/AUD one week later.

- By what percentage did the Australian dollar appreciate or depreciate against the dollar?
- By what percentage did the U.S. dollar appreciate or depreciate against the Australian dollar? (Hint: first express the rates as AUD/USD.)
- Why are your answers to (a) and (b) not exactly the same size?

5. (8 points) Cross rates. Use the following quotes against the U.S. dollar: USD1.16/EUR, USD0.655/AUD, CNY7.12/USD, SEK9.40/USD.

- Calculate the cross rate between the Australian dollar and the euro, in AUD/EUR and in EUR/AUD.
- Calculate the yuan–euro cross rate (CNY/EUR).
- Calculate the Swedish krona–Australian dollar cross rate (SEK/AUD).
- Explain in one or two sentences why the dollar is called a “vehicle currency.”

6. (7 points) Bid–ask spreads in the interbank market. A dealer quotes the British pound at USD1.3318–1.3322/GBP.

- Which rate is the bid and which is the ask? Which rate do you get if you buy pounds from the dealer? If you sell pounds to the dealer?
- What is the spread in pips? What is the spread as a percentage of the ask price?
- A company buys GBP2,000,000 from the dealer and, after changing its mind, sells the pounds back to the dealer a minute later at the same quotes. How many dollars does the company lose?

7. (6 points) Bid–ask spreads at the airport. At the airport, a currency kiosk buys pounds at USD1.25/GBP and sells pounds at USD1.40/GBP.

- What is the percentage spread? Compare it with your answer to Problem 6(b).
- You exchange \$1,500 into pounds at the kiosk, then change your mind and exchange all the pounds back into dollars. How many dollars do you end up with?
- Give two reasons why spreads for cash (“physical exchange”) are so much larger than interbank spreads.

8. (5 points) Flipping a bid–ask quote. A bank quotes the dollar at CHF0.7995–0.8005/USD. What are the bank's bid and ask prices for the Swiss franc, quoted in USD/CHF? (Hint: remember that the bank always buys low and sells high.)

9. (6 points) Cross rates with bid–ask spreads. A bank quotes USD1.3318–1.3322/GBP and CHF0.7995–0.8005/USD. A customer wants to trade pounds for Swiss francs, and the bank will go through the dollar.

- What is the bank's bid rate for pounds in terms of francs (CHF/GBP)?
- What is the bank's ask rate for pounds in terms of francs (CHF/GBP)?

10. (6 points) A Swiss watch company must pay GBP750,000 to a British supplier. It uses the bank and the quotes in Problem 9.

- Which CHF/GBP rate applies, and how many Swiss francs will the payment cost?
- How many more francs does the company pay than it would at the mid-rates of both quotes?

11. (7 points) Cross rates over time. On September 1, 2026 the quotes were USD1.1500/EUR and CNY7.1500/USD. On October 1, 2026 they were USD1.1700/EUR and CNY7.1000/USD.

- Calculate the CNY/EUR cross rate on both dates.
- By what percentage did the euro appreciate or depreciate against the yuan?
- By what percentage did the yuan appreciate or depreciate against the euro?

12. (7 points) As a foreign exchange trader, you see the following quotes for Australian dollars (AUD), U.S. dollars (USD), and Japanese yen (JPY):

USD0.6550/AUD JPY99.20/AUD JPY148.60/USD

Is there an arbitrage opportunity, and if so, how would you exploit it? Compute your profit per \$1,000,000.

13. (10 points) **Triangular arbitrage.** You observe the following quotes (ignore bid–ask spreads): USD0.6550/AUD in New York, USD0.7250/CAD in Toronto, and CAD0.9200/AUD in Sydney.

- What CAD/AUD cross rate is implied by the two dollar quotes?
- Is there an arbitrage opportunity? In which market is the Australian dollar “too expensive”?
- You start with \$1,000,000. Describe the three trades you would make and calculate your profit in dollars.
- What will happen to the three exchange rates as many traders do the same thing?

14. (10 points) **Triangular arbitrage with bid–ask spreads.** Now the quotes are:

USD0.6548–0.6552/AUD, USD0.7248–0.7252/CAD, and CAD0.9020–0.9030/AUD. Starting with \$1,000,000, check both possible routes (dollars → Australian dollars → Canadian dollars → dollars, and dollars → Canadian dollars → Australian dollars → dollars). Is there an arbitrage profit once you pay the bid–ask spreads?

15. (6 points) **The foreign exchange market.** Answer briefly.

- What is “liquidity,” and why do the most-traded currency pairs (such as EUR/USD) have smaller bid–ask spreads than rarely traded pairs (such as SEK/MXN)?
- What is the interbank market, and how does it differ from the retail market in which tourists and small firms buy currency?
- A dealer does not try to predict exchange rates. How does the dealer still make a profit?